Nurul Nazihah Jamal

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OBJECTIVE

Dedicated educator and researcher with a strong academic background in software engineering and informatic engineering, specializing in adaptive learning frameworks, competency modelling, and educational technology integration. Experienced in higher education teaching, and curriculum development. Proven track record of publishing in indexed journals, leading and engaging in community outreach. Committed to advancing academic excellence through impactful teaching, innovative research, and meaningful community service. My objective is to always be able in giving good benefits to everyone in whatever shape they may take as I believed that no one has ever become poor by giving.

EDUCATION

Universiti Teknologi Malaysia, Skudai Johor **Skudai, Johor, Malaysia**
Doctor of Philosophy (Informatic Engineering) *(Mar 2022 – Aug 2025)*

- Thesis: An Ontology-Driven Learning Analytics Framework for Diagnosing and Predicting Students' Competencies

Universiti Teknologi Malaysia, Skudai Johor **Skudai, Johor, Malaysia**
Master of Philosophy (Software Engineering) *(Sept 2019 – Nov 2021)*

- Thesis: Adaptive Learning Model for Learning Computational Thinking through Educational Robotics

Universiti Teknologi Malaysia, Skudai Johor **Skudai, Johor, Malaysia**
Bachelor of Computer Science (Software Engineering) *(Aug 2015-Aug 2019)*

- Dissertation: Reusability of Real-Time Bus Tracking System for Multiple Platform
- CGPA: 3.74

ACADEMIC APPOINTMENT & TEACHING EXPERIENCE

Assistant Fellow Kolej Tun Hussein Onn **Skudai, Johor, Malaysia**
Universiti Teknologi Malaysia, Skudai Johor *(Sept 2019 – Apr 2025)*

- *Coordinated student engagement activities, academic mentoring, and leadership programmes.*

Teaching Assistant/Partime Lecturer **Skudai, Johor, Malaysia**
Faculty of Computing *(2024 – 2025)*

- Conducted tutorials, supervised projects, and supported lab sessions for two selected subjects.

School of Professional and Continuing Education *(2024 – 2025)*

- Assisted computing-related training for continuing education programmes for two selected subjects.

Kuala Nerus, Terengganu, Malaysia

Faculty of Computer Science and Mathematics *(2025-Recent)*

- Assisted computing-related teaching and training for two selected subjects.

PROFESSIONAL CERTIFICATION

- Certified Professional for Requirement Engineering (CPRE), 2017
- Professional Certificate for Internal Halal Auditor, 2022
- QuickHalal Certified Professionals, 2022

RESEARCH PUBLICATION

Journal Articles

- **Jamal, Nurul N.,** Dayang NA Jawawi, Rohayanti Hassan, Radziah Mohamad, Shahliza A. Halim, Nor A. Saadon, Mohd A. Isa, and Haza NA Hamed. "Adaptive learning framework for learning computational thinking using educational robotics." *Computer Applications in Engineering Education* 32, no. 4 (2024): e22732. (***Web Of Science Indexed – Q2***)
- Hui, Sung Sheau, Dayang NA Jawawi, and **Nurul N. Jamal.** "Integration of Adaptive Game-Based Learning Approach in Learning Mathematics Subject for Primary School." *International Journal of Innovative Computing* 14, no. 1 (2024): 43-47. (***MyCite Indexed***)
- Jawawi, Dayang, **Nurul Jamal,** Shahliza Abdul Halim, Nor Sa'adon, Rosbi Mamat, Mohd Isa, Radziah Mohamad, and Haza Abdull Hamed. "Nurturing secondary school student computational thinking through educational robotics." *International Journal of Emerging Technologies in Learning (iJET)* 17, no. 3 (2022): 117-128. (***Scopus Indexed***)
- **Jamal, Nurul Nazihah,** Dayang Norhayati Abang Jawawi, Rohayanti Hassan, and Rosbi Mamat. "Conceptual model of learning computational thinking through educational robotic." *International Journal of Emerging Technologies in Learning (Online)* 16, no. 15 (2021): 91. (***Scopus Indexed***)

Conference Proceedings

- Jawawi, Dayang NA, Shahliza Abdul Halim, Nadirah Mohamad, **Nurul N. Jamal**, Nor Azizah Saadon, Radziah Mohamad, Haza Nuzly Abdull Hamed, and Mohd Adham Isa. "Collaborative learning approach for teaching computational thinking using mobile robot during COVID19 pandemic." In AIP Conference Proceedings, vol. 3056, no. 1, p. 080002. AIP Publishing LLC, 2025.
- **Jamal, Nurul N.,** and Dayang NA Jawawi. "Review on the initiatives taken by the education sectors to nurture CT with ubiquitous learning paradigm." In AIP Conference Proceedings, vol. 3056, no. 1, p. 060001. AIP Publishing LLC, 2025. (***Scopus Indexed***)
- Jawawi, Dayang NA, Nor Azizah Saadon, **Nurul Nazihah Jamal,** Shahliza Abd Halim, Radziah Mohamad, Mohd Adham Isa, and Haza Nuzly Abdul Hamed. "Fostering Programming Resilience: Developing a Scale for Secondary Schools." In 2024 IEEE 13th International Conference on Engineering Education (ICEED), pp. 1-6. IEEE, 2024. (***Scopus Indexed***)
- **Jamal, Nurul N.,** Dayang NA Jawawi, Roliana Ibrahim, and Alif Ridzuan Khairudin. "Development of InCLOOM: A Competency Ontology Model." In 2024 IEEE 13th International Conference on Engineering Education (ICEED), pp. 1-6. IEEE, 2024. (***Scopus Indexed***)
- **Jamal, Nurul Nazihah**, Dayang Norhayati Abang Jawawi, Rohayanti Hassan, and Rosbi Mamat. "Conceptual model of computational thinking through educational robotics." In *AIP Conference Proceedings*, vol. 2433, no. 1, p. 020014. AIP Publishing LLC, 2022. (***Scopus Indexed***)

- **Jamal, Nurul N.,** Dayang NA Jawawi, Rohayanti Hassan, and RM Irsyad Kamil. "Adaptive learning in computing education: A systematic mapping study." In IOP Conference Series: Materials science and engineering, vol. 864, no. 1, p. 012069. IOP Publishing, 2020. (***Scopus Indexed***)
- Kamil, R. M. Irsyad, Dayang NA Jawawi, Rosbi Mamat, and **Nurul N. Jamal**. "Indoor Smart Wheelchair: Systematic Mapping." In IOP Conference Series: Materials Science and Engineering, vol. 864, no. 1, p. 012070. IOP Publishing, 2020. (***Scopus Indexed***)
- Sharif, Shafri A., Mohd S. Suhaimi, **Nurul N. Jamal**, Irsyad K. Riadz, Ikmal F. Amran, and Dayang NA Jawawi. "Real-Time campus university bus tracking mobile application." In 2018 Seventh ICT International Student Project Conference (ICT-ISPC), pp. 1-6. IEEE, 2018. (***Scopus Indexed***)

NEWS & OUTREACH ARTICLES [as a writer]

- **UTM Leads RoboKar CSR Program, Strengthening Educational Bonds in Pagar Alam** – *UTM NewsHub, 6 Nov 2024.*
- **UTM Santuni Pelajar Sekolah Menengah Daerah Segamat dengan Kemahiran Asas AI bersama Mahasiswa UNILED, Indonesia** – *UTM NewsHub, 16 May 2024.*
- **Dr. Eleen Dayana Dinobat Penerima Anugerah Canselor dalam MK66** – *UTM NewsHub, 25 Nov 2022.*
- **"Prayer, Patience and Hardwork", Three Keys to Survive PhD** – *UTM NewsHub, 24 Nov 2022.*
- **Sehati Sejiwa, Bergraduasi Bersama** – *UTM NewsHub, 24 Nov 2022.*
- **Walau Perit, Anisya Cekal Tamatkan Pengajian dan Peroleh Ijazah Kelas Pertama** – *UTM NewsHub, 23 Nov 2022.*
- **Pasangan Kembar Raih Ijazah Sarjana Muda, Kini Sama-sama Jadi Jurutera** – *UTM NewsHub, 21 Nov 2022.*

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- Microcredential UTM - Modul pemikiran komputasional untuk penyelesaian masalah menggunakan pengaturcaraan robotik. *By ICC - IP/CR/02574*
- Microcredential UTM – Video Animasi Bagi Pemikiran Komputasional Untuk Penyelesaian Masalah Menggunakan Pengaturcaraan Robotik. *By ICC - IP/CR/02394*

PARTICIPATION IN ACADEMIC ACTIVITIES

- Presenter, *Sustainable & Integrated Engineering International Conference SIE (SIE 2022)*
- Presenter, *Virtual International Conference on Ubiquitous Education and Learning (VICUEL 2020)*
- Presenter, *International Conference on Applied Computing (ICAC 2020, 2023)*
- Committee, *Postgraduate Annual Research Seminars (PARS 2023)*
- Presenter, *International Conference on Engineering Education (ICEED 2025)*
- Participant, *New Academia Learning Innovation (NALI 2021, 2023, 2024) Exhibition*
- Participant, *Pars2024 Keynote Session: The Malaysian Social Media Sentiment Analysis Challenge - How Ai Helps (2024)*
- Speaker, *Sharing Session Mafan (2024)*

PARTICIPATION IN EXTRA-CURRICULAR ACTIVITIES

- Speaker, *Program Bengkel Pembuatan Sijil Online (2024)* - University

- Committee, *International DigiCulture Coding CSR Programme: RoboKar AI Programming (2024)* - International
- Committee, *Colloqium and Transfer Knowledge ITS, Bandung, (2023)* - International
- Committee, *Bengkel Penyelesaian Masalah melalui Pengaturcaraan RoboKar, (2019-2024)* - National
- Committee, *Robotic Programming Workshop, (2019-2024)* – University & National
- Participant, *Kursus Induksi Felo UTM, (2023)* – University
- Committee, *Ktho River Rangers: Pulihkan Ekosistem Alam, (2025)* – National
- Speaker, *Program Perancangan Pembelajaran Pemikiran Komputasional Melalui Pengaturcaraan Robot Pejabat Pendidikan Daerah Segamat Bersama Universiti Teknologi Malaysia, (2021)* – National
- Speaker, *International Digiculture Coding CSR Programme: Robokar Ai Programming, (2024)* - International

AWARDS

- Merit Thesis Award PhD (2025)
- NALI 2024 Award Competition (2024)
- Anugerah Pengembangan Ilmu Muda (2024)
- Best Presenter & Best Paper for Sustainable & Integrated Engineering International Conference (SIE 2022)
- Anugerah Figura Muda IPF (2019)
- Dean List Receiver (2019)

SKILLS & INTERESTS

Research Interests: Engineering Education, Applied Soft Computing, Data Analytics

Software Application: Microsoft Office Suite, GitHub, WordPress, Streamlit

Programming Language: C++, Java, HTML, CSS, PHP, Python

REFERENCES

[Upon Request]